

AI Agents and Agentic AI with Python & Generative AI

Module 1

Agentic AI Concepts

- ✓ Introduction
Video • 10 min
- ✓ Flipped Interaction Pattern
Video • 7 min
- ✓ The Agent Loop
Video • 8 min
- ✓ Knowledge Check: Agent Loop & Flipped Interaction
Practice Assignment • Grade: 100%
- ✓ Running the Code Samples in the Course
Reading • 10 min
- ✓ Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- ✓ Try Out Programmatic Prompting
Reading • 10 min
- ✓ Programmatic Prompting for Agents II
Ungraded Plugin • 15 min
- ✓ Programmatic Prompting for Agents III
Ungraded Plugin • 15 min
- Programmatic Prompting: Building Blocks for AI Agents
Dialogue • 15 min
- ✓ Try Out the Customer Service Agent
Reading • 10 min
- ✓ Giving Agents Memory
Ungraded Plugin • 15 min
- ✓ Practicing Programmatic Prompting for Agents
Ungraded Plugin • 15 min
- Practicing Programmatic Prompting for Agents (Solution)
Ungraded Plugin • 15 min
- ✓ Adding Structure to AI Agent Outputs
Video • 14 min
- ✓ Building Your First Agent
Ungraded Plugin • 15 min
- ✓ AI Agent / Environment Interface
Ungraded Plugin • 15 min
- ✓ AI Agent Feedback and Memory
Ungraded Plugin • 15 min
- ✓ The Agent Loop: From Understanding to Thinking Like an Agent
Dialogue • 15 min
- ✓ Learning More & Staying Connected
Reading • 10 min
- Understanding Agentic AI Concepts
Graded Assignment • 30 min

Module 2

AI Agents, Tools, Actions, & Language

Module 3

GAME: A Conceptual Framework for AI Agents

Module 4

Agent Tool Mangement

Learning More & Staying Connected

00 0:30 / 2:20 2x

If you haven't taken my first course, [Prompt Engineering for ChatGPT](#), it is a great complement to this course that also teaches concepts that every prompt engineer can benefit from..

Stay Up to Date in a Rapidly Changing AI Landscape — Join the Private Community for Professionals in My Coursera Classes

If you're taking any of my Coursera courses, you're invited to join our private Circle community! While the courses give you core skills and frameworks, this space is for going beyond the syllabus—with extra resources, office hours, monthly updates on key trends in AI, and more. It also includes exclusive content that enriches and extends topics from all 20+ of my courses — and it's only available to professionals actively taking my Coursera courses. It's the best way to stay current, go deeper, and connect with others on the same professional learning journey.

Join the Community [here](#)

Connect with Me on LinkedIn

If you would like to stay connected to what we are doing at Vanderbilt as part of the Initiative on the Future of Learning & Generative AI, please don't hesitate to reach out and connect with me on [LinkedIn](#).

Learning More About Prompt Engineering

After this course (or concurrently), you can learn more about using ChatGPT and how to write effective prompts in my course on [Prompt Engineering for ChatGPT](#). Here are some other courses you can check out:

- Learn More About Custom GPTs [here](#)
- Learn More About ChatGPT Advanced Data Analysis [here](#)
- Learn how specifically Generative AI is going to transform computing and industry in [my Generative AI Primer](#) course

All My Generative AI Courses are Here

I teach a number of additional courses on Generative AI, including: GPT-4 Vision, Innovative Teaching with ChatGPT, ChatGPT Advanced Data Analysis, Generative AI Primer, Generative AI for Leaders, Trustworthy Generative AI, and more. [You can see all the Generative AI courses that I teach here.](#)

Learn More Prompt Patterns in Our Paper:

- [A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT](#) by Jules White, Quichen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, Douglas C. Schmidt
- Read a Paper on "[Living Software Systems](#)" and the future that AI Agents will make possible.

Go to next item

✓ Completed

Like Dislike Report an issue

Go to next item →